UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
**United States Patent and Trademark Office**
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO. | CONFIRMATION NO. |
|---|---|---|---|---|
| 17/703,514 | 03/24/2022 | Yu Fang | SONA-0006-U01-C05 | 2906 |

87084 7590 07/21/2023

GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

| EXAMINER |
|---|
| ROY, SANJOY K |

| ART UNIT | PAPER NUMBER |
|---|---|
| 2443 | |

| NOTIFICATION DATE | DELIVERY MODE |
|---|---|
| 07/21/2023 | ELECTRONIC |

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

| Office Action Summary | Application No. 17/703,514 | Applicant(s) Fang et al. | |
|---|---|---|---|
| | Examiner SANJOY K ROY | Art Unit 2443 | AIA (FITF) Status Yes |

*-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --*

**Period for Reply**

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

**Status**

1)☑ Responsive to communication(s) filed on 03/24/2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

**Disposition of Claims***

5) ☑ Claim(s) 1-25 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-25 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

**Application Papers**

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

**Priority under 35 U.S.C. § 119**

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

**Certified copies:**

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

**Attachment(s)**

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 03/24/2022, 03/24/2022.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

## DETAILED ACTION

### *Notice of Pre-AIA or AIA Status*

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. Claims 1-25 have been examined and rejected.

### *Double Patenting*

3 The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the claims at issue are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., In re Berg, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); In re Goodman, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); In re Longi, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); In re Van Ornum, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); In re Vogel, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and In re Thorington, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the reference application or patent either is shown to be commonly owned with this application, or claims an invention made as a result of

activities undertaken within the scope of a joint research agreement. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO internet Web site contains terminal disclaimer forms which may be used. Please visit http://www.uspto.gov/forms/. The filing date of the application will determine what form should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to http://www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

4. Claims 1-25 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-34 of co-pending U.S. Application No. 17570018. Although the claims at issue are not identical, they are not patentably distinct from each other because claims in the co-pending application anticipated claims in the present application.

| Instant Application 17703514 | US Application No. 17570018 |
|---|---|
| 1. An apparatus comprising: a first network interface circuit structured to interpret a first Ethernet data set and to transmit a second Ethernet data set; a second network interface circuit structured to transmit a first controller area network (CAN) data set and to interpret a second CAN data set;<br><br>and a translation circuit structured to: determine a first message value from the first Ethernet data set and to encapsulate the first message value in the first CAN data set; and determine a second message value from the | 18. An apparatus comprising: a first network interface circuit structured to interpret a first network data set having a first network format;<br><br>a translation circuit structured to determine a message value from the first network data set and to encode the message value in a second network data set having a second network format; |

| | |
|---|---|
| second CAN data set and to encapsulate the second message value in the second Ethernet data set; | and a second network interface circuit structured to transmit the second network data set; |
| | wherein: the first network format and the second network format are vehicle data formats; and the first network format is different than the second network format. |
| | 23. The apparatus of claim 18, wherein one of the first network format or the second network format is Controller Area Network (CAN) based, and the other of the first network format or the second network format is Ethernet based. |
| | 24. The apparatus of claim 23, wherein one or more CAN messages of the CAN based network can be carried in one Ethernet frame of the Ethernet based network. |
| wherein the first network interface circuit, the translation circuit, and the second network interface circuit are defined by an Ethernet switch and/or a CAN gateway both structured to be incorporated into a mobile system. | 26. The apparatus of claim 18, wherein the first network interface circuit and/or the second network interface circuit is integrated with an Ethernet switch. |

The scope of independent Claims 12 and 20 of the instant application is covered by claims 1 and 28 and related dependent claims of the co-pending application 17570018.

### *Claim Rejections - 35 USC § 112*

5. The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

6. Claims 20-25 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

As per independent claim 20

The claim terms " the CAN gateway" is unclear as it is not clear if "the CAN gateway" refers to "a converged network device (CND)" or "a configurable edge gateway (CEG)" or completely different gateway.

### *Claim Rejections - 35 USC § 102*

7. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

8. Claims 1-14 and 17-25 are rejected under AIA 35 U.S.C. 102(a)(1) as being anticipated by Rajan et al. (U.S. PGPub 2019/0141133).

Rajan teaches an apparatus (Rajan, see para 0044, the in-vehicle gateway 136 handles both intra-protocol switching and inter-protocol switching CAN-to-CAN traffic as well as CAN-to-Ethernet traffic) comprising:

a first network interface circuit structured to interpret a first Ethernet data set (Rajan, see para 0051 shown in fig 11 an Ethernet frame arrives at the Ethernet switch 804 at any of the Ethernet interface port 806 with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, at 4, the

CPU 814 reads the encapsulation information and performs a routing function and determines the CAN bus for the message identifier, and at 5 passes the Ethernet frame down the software protocol stack 604 converting the Ethernet frame to a CAN message protocol format, and sends the CAN message to the CAN controller for transmission)

and to transmit a second Ethernet data set (Rajan see para 0049 as shown in fig. 10, pre-configured Ethernet header encapsulation from the translation table is pre-pended to the translated CAN message to generate a Ethernet message payload sent the Ethernet switch 804 via the MAC 810 to be transmitted using the ethernet interface ports 806 and 808);

a second network interface circuit structured to transmit a first controller area network (CAN) data set (Rajan, see para 0051 shown in fig 11 at 5 when the Ethernet frame is converted to a CAN message protocol format the CAN controller transmits CAN message using the interface 138)

and to interpret a second CAN data set (Rajan, see para 0048, 0049 as shown in fig. 10, as shown in fig. a CAN message arrives at the CAN controller using the CAN interface 138 hash circuitry 826 determines an index value for the translation table, to covert a 29-bit message identifier to an 11-bit index value, the CPU uses the message identifier to perform a routing function and determine that the message destination in this example is an Ethernet port, using the 11-bit hash or received message identifier look up look up the corresponding pre-configured Ethernet header encapsulation from the translation table and pre-pended the pre-configured header to the message payload to be sent the Ethernet switch 804 via the MAC 810);

and a translation circuit structured to: determine a first message value from the first Ethernet data set and to encapsulate the first message value in the first CAN data set (Rajan, see para 0051, as shown in fig. 11 Ethernet to CAN message flow at 3, the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, the CPU 814 reads the encapsulation information and performs a routing function and determines the destination, CAN bus for the message identifier, a 5, the software protocol stack 604, which converts the Ethernet frame to a CAN message protocol format, and sends the CAN message to the CAN controller for transmission);

and determine a second message value from the second CAN data set and to encapsulate the second message value in the second Ethernet data set (Rajan, see para 0048, 0049 as shown in fig. 10, as shown in fig. a CAN message arrives at the CAN controller using the CAN interface 138 hash circuitry 826 determines an index value for the translation table, to covert a 29-bit message identifier to an 11-bit index value, the CPU uses the message identifier to perform a routing function and determine that the message destination in this example is an Ethernet port, using the 11-bit hash or received message identifier look up look up the corresponding pre-configured Ethernet header encapsulation from the translation table and pre-pended the pre-configured header to the message payload to be sent the Ethernet switch 804 via the MAC 810);

wherein the first network interface circuit, the translation circuit, and the second network interface circuit are defined by an Ethernet switch and/or a CAN gateway both

structured to be incorporated into a mobile system (Rajan, see para 0042, 0044 in-vehicle network gateway 136 handles both intra-protocol switching and inter-protocol switching, CAN-to-CAN traffic as well as CAN-to-Ethernet traffic, the in-vehicle network gateway 136 may use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to the CAN interface, the CPUs 814 may implement the control circuitry 214 with RAM 816 implementing the memory system and store the routing table 211, translation table 212, pre-defined headers 218-1-218-*n*, the software protocol stack 604, buffers 822 for storing messages, message filters 824, the hash circuitry 826, and the routing tables 828 for determining the destination interface, as an alternative implementation to routing tables stored in the RAM 816).

As per claim 2

Rajan teaches the apparatus of claim 1 further comprising a configuration circuit defined by the Ethernet switch and/or the CAN gateway and structured to modify at least one of the first network interface circuit, the translation circuit, or the second network interface circuit in response to a configuration command value (Rajan see para 0035, 0036, the optional Hardware acceleration is enabled or disabled according to a global configuration parameter, or enabled or disabled on a message-by-message basis, in configuration settings stored in memory in-vehicle network gateway 136 implements a bypass mode of hardware acceleration, the control circuitry 214 is coupled to a memory system 602 with a software protocol stack 604 and configuration settings 606 for the in-vehicle network gateway 136, configuration

settings 606 may be set by any authorized system operator indicating whether the in-vehicle network gateway 136 will execute hardware acceleration for protocol translation, including message filtering, pre-defined header lookup, and message aggregation, these three acceleration aspects is enabled or disabled separately within the configuration settings 606).

As per claim 3

Rajan teaches the apparatus of claim 2, wherein the configuration circuit is further structured to selectively configure one or more portions of the first network interface circuit, the translation circuit, and/or the second network interface circuit to be defined by the Ethernet switch and/or the CAN gateway (Rajan see para 0036, 0039, in-vehicle network gateway 136 will execute hardware acceleration for protocol translation, including message filtering, pre-defined header lookup, and message aggregation, these three acceleration aspects is enabled or disabled separately within the configuration settings 606, message filter 210 determines which of the messages to pass for hardware accelerated protocol translation and which to discard or ignore, the translation table 212 defined for the Ethernet interface has an index input that receives a table index value and for each value of the index input, stores a pointer to a pre-defined Ethernet header).

As per claim 4

Rajan teaches the apparatus of claim 1, wherein the first network interface circuit is defined by the Ethernet switch incorporated into the mobile system and the second

network interface circuit is defined by the CAN gateway incorporated into the mobile system (Rajan see para 0041, the gateway circuitry 802 includes an Ethernet switch 804 with ports 806 and 808. The Ethernet switch 804 connects through Media Access Control circuitry MAC 810 to a system bus 812, bus transceiver circuitry 820 includes CAN bus transceivers connected to the CAN interface 138).

As per claim 5

Rajan teaches the apparatus of claim 4, wherein the translation circuit is defined by the CAN gateway incorporated into the mobile system (Rajan see para 0042, the CPUs 814 implement all or part of the control circuitry 214 , while the RAM 816 implement the memory system and store the routing table 211, translation table 212, pre-defined headers 218-1-218-*n*, the software protocol stack 604, and provide general purpose memory storage for the in-vehicle network gateway 136, the hardware acceleration circuitry 818 include, buffers 822 for storing messages, message filters 824, the hash circuitry 826, and the routing tables 828 for determining the destination interface as an alternative implementation to routing tables stored in the RAM 816).

As per claim 6

Rajan teaches the apparatus of claim 4, wherein the translation circuit is defined by the Ethernet switch incorporated into the mobile system (Rajan see para 0044, 0051the in-vehicle network gateway 136 use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to

the CAN or other interface, the gateway circuitry 802 shapes the switch queues to match the rate of the target network or bus interface. At (3), the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame).

As per claim 7

Rajan teaches the apparatus of claim 1, wherein the translation circuit is further structured to determine a priority of a flow related to the second CAN data set, and to apply the priority of the flow to the second Ethernet data set (Rajan see para 0032, 0044, the in-vehicle network gateway 136 may provide priority handling of messages, prioritize high priority messages over low priority message, the in-vehicle network gateway 136 may treat non-Ethernet interfaces as logical ports, and add different queues per non-Ethernet interface assigned to low and high priority traffic, CAN bus messages flowing to the Ethernet network can be aggregated by the control circuitry 214 by aggregating the message payloads with additional subsequently received message payloads into a single Ethernet packet, multiple message payloads associated with a given specific protocol, the CAN bus protocol may be destined for a common recipient, as reflected by having a common pre-defined header determined from the individual message identifiers as received, the control circuitry 214 may aggregate these message payloads into a common outgoing message by performing aggregation, until a transmission criterion or criteria are satisfied such as a message payload with greater than a pre-defined threshold level of priority is received, and this preempt low priority message aggregation).

As per claim 8

Rajan teaches the apparatus of claim 1, wherein the translation circuit is further structured to determine a priority of the second CAN data set, and to apply the priority to the second Ethernet data set (Rajan see para 0032, 0044, the in-vehicle network gateway 136 may provide priority handling of messages, prioritize high priority messages over low priority message, the in-vehicle network gateway 136 may treat non-Ethernet interfaces as logical ports, and add different queues per non-Ethernet interface assigned to low and high priority traffic, CAN bus messages flowing to the Ethernet network can be aggregated by the control circuitry 214 by aggregating the message payloads with additional subsequently received message payloads into a single Ethernet packet, multiple message payloads associated with a given specific protocol, the CAN bus protocol may be destined for a common recipient, as reflected by having a common pre-defined header determined from the individual message identifiers as received, the control circuitry 214 may aggregate these message payloads into a common outgoing message by performing aggregation, until a transmission criterion or criteria are satisfied such as a message payload with greater than a pre-defined threshold level of priority is received, and this preempt low priority message aggregation).

As per claim 9

Rajan teaches the apparatus of claim 1, wherein the second CAN data set comprises a service oriented communication, and wherein the translation circuit is further structured to mark the second Ethernet data set as a service oriented

communication (Rajan see para 0049 the outgoing message is queued as an Ethernet frame in the appropriate queue at the egress port or ports, the queue may be selected responsive to class of service information present in the outgoing message and specified, in the pre-configured header and detected by processing circuitry in the ingress pipeline of the Ethernet switch 804).

As per claim 10

Rajan teaches the apparatus of claim 9, wherein the translation circuit is further structured to apply a priority of the service oriented communication to the second Ethernet data set (Rajan see para 0032, 0044, the in-vehicle network gateway 136 may provide priority handling of messages, prioritize high priority messages over low priority message, the in-vehicle network gateway 136 may treat non-Ethernet interfaces as logical ports, and add different queues per non-Ethernet interface assigned to low and high priority traffic, CAN bus messages flowing to the Ethernet network can be aggregated by the control circuitry 214 by aggregating the message payloads with additional subsequently received message payloads into a single Ethernet packet, multiple message payloads associated with a given specific protocol, the CAN bus protocol may be destined for a common recipient, as reflected by having a common pre-defined header determined from the individual message identifiers as received, the control circuitry 214 may aggregate these message payloads into a common outgoing message by performing aggregation, until a transmission criterion or criteria are satisfied such as a message payload with greater than a pre-defined threshold level of priority is received, and this preempt low priority message aggregation).

As per claim 11

Rajan teaches the apparatus of claim 1, wherein the translation circuit is further structured to determine a CAN identifier of the second CAN data set, and to apply a priority to the second Ethernet data set in response to the CAN identifier(Rajan see para 0048, 0049 the CPU uses the message identifier generated from hash circuitry 826 as an index value for the translation table converting a 29-bit message identifier to an 11-bit index value to perform a routing function and determine that the message destination in this example is an Ethernet port, at the index value the 11-bit hash or received message identifier is applied to the translation table to look up the corresponding pre-configured Ethernet header to pre-pended to the message payload and the resulting outgoing message is sent to the Ethernet switch 804 via the MAC 810 with aggregation of multiple incoming message payloads placed into a single outgoing Ethernet packet that is queued as an Ethernet frame in the appropriate queue at the egress port selected responsive to class of service information present in the outgoing message and specified in the pre-configured header and detected by processing circuitry in the ingress pipeline of the Ethernet switch 804, the outgoing message the Ethernet switch 804 schedules and transmits the outgoing message out of the destination port based on the priority of the queues set by the class service information).

As per claim 12

Rajan teaches the method (Rajan, see para 0044, in-vehicle network gateway 136 handles both intra-protocol switching and inter-protocol switching CAN-to-

CAN traffic as well as CAN-to-Ethernet traffic represented as a converged network device) comprising:

interpreting, via a first interface circuit of a converged network device (CND), a first Ethernet data set (Rajan, see para 0051 shown in fig 11 an Ethernet frame arrives at the Ethernet switch 804 at any of the Ethernet interface port 806 with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, at 4, the CPU 814 reads the encapsulation information and performs a routing function and determines the CAN bus for the message identifier, and at 5 passes the Ethernet frame down the software protocol stack 604 converting the Ethernet frame to a CAN message protocol format, and sends the CAN message to the CAN controller for transmission);

determining, via a translation circuit of the CND, a first message value from the first Ethernet data set (Rajan, see para 0051, as shown in fig. 11 Ethernet to CAN message flow at 3, the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, the CPU 814 reads the encapsulation information and performs a routing function and determines the destination, CAN bus for the message identifier);

encapsulating, via the translation circuit, the first message value in a first controller area network (CAN) data set (Rajan see para 0027, 0051, at step 5 the CPU passes the Ethernet frame down the software protocol stack 604, which converts the Ethernet frame to a CAN message protocol format by, translation table 212 in RAM 816 may provide a lookup to map the received message to a pre-defined headers of an

outgoing CAN message using pointers 216 to any number of pre-defined headers 218-1 through 218-*n*, stored in the RAM) ;

transmitting, via a second interface circuit of the CND, the first CAN data set (Rajan, see para 0051 shown in fig 11 at 5 when the Ethernet frame is converted to a CAN message protocol format the CAN controller transmits CAN message using the interface 138);

interpreting, via the second interface circuit, a second CAN data set (Rajan, see para 0047 as shown in fig. 10 CAN to Ethernet message flow, at 1 a CAN message arrives at the CAN controller through the interface 138 and is stored in the buffer 822, at 2 the message filters 824 apply to pass messages of interest to the CPU for further processing, the hash circuitry 826 determines an index value for the translation table, to covert a 29-bit message identifier to an 11-bit index value, if the length of the received message identifier is already compatible with the translation table address space, the hash circuitry 826 is skipped).

determining, via the translation circuit, a second message value from the second CAN data set (Rajan, see para 0048 as shown in fig. 10, the CPU uses the CAN message identifier to perform a routing function and determine that the message destination is an Ethernet port, using the 11-bit hash or received message identifier look up the corresponding pre-configured Ethernet header encapsulation from the translation table);

encapsulating, via the translation circuit, the second message value in a second Ethernet data set (Rajan see para 0049 at 6 the retrieved pre-configured header is pre-pended to the message payload and the resulting outgoing message is sent to the

Ethernet switch 804 via the MAC 810, if aggregation is enabled, the aggregation may result in multiple incoming message payloads placed into a single outgoing Ethernet packet);

and transmitting, via the first interface circuit, the second Ethernet data set (Rajan see para 0049 at 7 the outgoing message is queued as an Ethernet frame in the appropriate queue at the egress port or ports, at 8 the outgoing message the Ethernet switch 804 schedules and transmits the outgoing message out of the destination port);

wherein the CND is disposed onboard a vehicle and defined at least in part by an Ethernet switch and/or a CAN gateway (Rajan see para 0044, in-vehicle network gateway 136 handles both intra-protocol switching and inter-protocol switching, CAN-to-CAN traffic as well as CAN-to-Ethernet traffic the in-vehicle gateway 136 may use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to the CAN or other interface).

As per claim 13

Rajan teaches the method of claim 12 further comprising: interpreting a configuration command value; and modifying the CND in response to the configuration command value (Rajan see para 0035, 0036, the optional Hardware acceleration is enabled or disabled according to a global configuration parameter, or enabled or disabled on a message-by-message basis, in configuration settings stored in memory in-vehicle network gateway 136 implements a bypass mode of hardware acceleration, the control circuitry 214 is coupled to a memory system 602 with a software protocol stack 604 and configuration settings 606 for the in-vehicle network gateway 136,

configuration settings 606 may be set by any authorized system operator indicating whether the in-vehicle network gateway 136 will execute hardware acceleration for protocol translation, including message filtering, pre-defined header lookup, and message aggregation, these three acceleration aspects is enabled or disabled separately within the configuration settings 606).

As per claim 14

Rajan teaches the method of claim 13, wherein modifying the CND comprises: selectively configuring which of one or more portions of the first interface circuit, the translation circuit and/or the second interface circuit are defined at least in part by the Ethernet switch and/or the CAN gateway (Rajan see para 0036, 0039, in-vehicle network gateway 136 will execute hardware acceleration for protocol translation, including message filtering, pre-defined header lookup, and message aggregation, these three acceleration aspects is enabled or disabled separately within the configuration settings 606, message filter 210 determines which of the messages to pass for hardware accelerated protocol translation and which to discard or ignore, the translation table 212 defined for the Ethernet interface has an index input that receives a table index value and for each value of the index input, stores a pointer to a pre-defined Ethernet header).

As per claim 17

Rajan teaches the method of claim 12, wherein the first network interface circuit is defined by the Ethernet switch and the second network interface circuit is defined by

the CAN gateway (Rajan see para 0041, the gateway circuitry 802 includes an Ethernet switch 804 with ports 806 and 808. The Ethernet switch 804 connects through Media Access Control circuitry MAC 810 to a system bus 812, bus transceiver 820 includes CAN bus transceivers connected to the CAN interface 138).

As per claim 18

Rajan teaches the method of claim 17, wherein the translation circuit is defined by the CAN gateway (Rajan see para 0042, the CPUs 814 implement all or part of the control circuitry 214 , while the RAM 816 implement the memory system and store the routing table 211, translation table 212, pre-defined headers 218-1-218-*n*, the software protocol stack 604, and provide general purpose memory storage for the in-vehicle network gateway 136, the hardware acceleration circuitry 818 include, buffers 822 for storing messages, message filters 824, the hash circuitry 826, and the routing tables 828 for determining the destination interface as an alternative implementation to routing tables stored in the RAM 816).

As per claim 19

Rajan teaches the method of claim 17, wherein the translation circuit is defined by the Ethernet switch (Rajan see para 0044, 0051the in-vehicle network gateway 136 use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to the CAN or other interface, the gateway circuitry 802 shapes the switch queues to match the rate of the target network

or bus interface at step 3 the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame).

As per claim 20

A system comprising: a vehicle comprising a first network zone comprising a Controller Area Network (CAN) based network, and a second network zone comprising an Ethernet based network (Rajan see para 0022, 0029 in the in-vehicle network gateway 136, there are multiple different automotive network or bus interfaces 138-146, including a protocol A automotive network or a CAN bus interface connected to a CAN network representing the first network zone with a address space of possible 11-bit message identifiers for CAN messages and a protocol B automotive network or an Ethernet interface connected to a Ethernet network as the second network zone, with a list of pre-configured ethernet header for each ethernet entity with Ethernet header data, source MAC address and destination MAC address, source and destination IP address, and TCP or UDP source and destination port, there may be an instance of buffer circuitry for each different network or bus interface 138-146);

a converged network device (CND) (Rajan, see para 0044, in-vehicle network gateway 136 handles both intra-protocol switching and inter-protocol switching CAN-to-CAN traffic as well as CAN-to-Ethernet traffic represented as a converged network device) comprising:

a first network interface circuit structured to interpret a first Ethernet data set and to transmit a second Ethernet data set (Rajan, see para 0051 shown in fig 11 an Ethernet frame arrives at the Ethernet switch 804 at any of the Ethernet interface port

806 with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, at 4, the CPU 814 reads the encapsulation information and performs a routing function and determines the CAN bus for the message identifier, and at 5 passes the Ethernet frame down the software protocol stack 604 converting the Ethernet frame to a CAN message protocol format, and sends the CAN message to the CAN controller for transmission);

a second network interface circuit structured to transmit a first CAN data set and to interpret a second CAN data set (Rajan, see para 0047 as shown in fig. 10 CAN to Ethernet message flow, at 1 a CAN message arrives at the CAN controller through the interface 138 and is stored in the buffer 822, at 2 the message filters 824 apply to pass messages of interest to the CPU for further processing, the hash circuitry 826 determines an index value for the translation table, to covert a 29-bit message identifier to an 11-bit index value, if the length of the received message identifier is already compatible with the translation table address space, the hash circuitry 826 is skipped);

and a translation circuit structured to: determine a first message value from the first Ethernet data set and to encapsulate the first message value in the first CAN data set (Rajan, see para 0051, as shown in fig. 11 Ethernet to CAN message flow at 3, the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, the CPU 814 reads the encapsulation information and performs a routing function and determines the destination, CAN bus for the message identifier);

and determine a second message value from the second CAN data set (Rajan see para 0048 as shown in fig. 10, the CPU uses the CAN message identifier to perform a routing function and determine that the message destination is an Ethernet port, using the 11-bit hash or received message identifier look up the corresponding pre-configured Ethernet header encapsulation from the translation table)

and to encapsulate the second message value in the second Ethernet data set (Rajan see para 0049 at 6 the retrieved pre-configured header is pre-pended to the message payload and the resulting outgoing message is sent to the Ethernet switch 804 via the MAC 810, if aggregation is enabled, the aggregation may result in multiple incoming message payloads placed into a single outgoing Ethernet packet);

an Ethernet switch comprising a portion of the Ethernet based network (Rajan see para 0041, the gateway circuitry 802 includes an Ethernet switch 804 with ports 806 and 808, Ethernet switch 804 connects through Media Access Control circuitry MAC 810 to a system bus 812, bus transceiver 820 includes CAN bus transceivers connected to the CAN interface 138);

a configurable edge gateway (CEG) comprising a portion of the CAN based network (Rajan see para 0035, 0036, 0041, the gateway circuitry 802 representing a configurable edge gateway includes an Ethernet switch 804 with ports 806 and 808, Ethernet switch 804 connects through Media Access Control circuitry MAC 810 to a system bus 812, system bus 812 provides a communication mechanism that interconnects the CPUs 814, the RAM 816, and the hardware acceleration circuitry 818, the MAC 810 performs bus conversion from the system bus 812 to the format of the Ethernet port interface at the Ethernet switch 804, bus transceiver circuitry 820 includes

CAN bus transceivers connected to the CAN interface 138 and connected to the system bus using the hardware acceleration circuitry 818, the optional Hardware acceleration is enabled or disabled according to a global configuration parameter, or enabled or disabled on a message-by-message basis, in configuration settings stored in memory in-vehicle network gateway 136 as the converged network device, connecting the CAN network to the Ethernet network, implements a bypass mode of hardware acceleration);

and wherein the first network interface circuit, the translation circuit, and the second network interface circuit are defined by the Ethernet switch and/or the CAN gateway (Rajan see para 0042, 0044 in-vehicle network gateway 136 handles both intra-protocol switching and inter-protocol switching, CAN-to-CAN traffic as well as CAN-to-Ethernet traffic, the in-vehicle network gateway 136 may use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to the CAN interface, the CPUs 814 may implement the control circuitry 214 with RAM 816 implementing the memory system and store the routing table 211, translation table 212, pre-defined headers 218-1-218-*n*, the software protocol stack 604, buffers 822 for storing messages, message filters 824, the hash circuitry 826, and the routing tables 828 for determining the destination interface, as an alternative implementation to routing tables stored in the RAM 816).

As per claim 21

Rajan teaches the system of claim 20 further comprising a configuration circuit defined by the Ethernet switch and/or the CEG and structured to modify the first network interface circuit, the translation circuit, and/or the second network interface circuit in

response to a configuration command value (Rajan see para 0035, 0036, the optional Hardware acceleration is enabled or disabled according to a global configuration parameter, or enabled or disabled on a message-by-message basis, in configuration settings stored in memory in-vehicle network gateway 136 implements a bypass mode of hardware acceleration, the control circuitry 214 is coupled to a memory system 602 with a software protocol stack 604 and configuration settings 606 for the in-vehicle network gateway 136, configuration settings 606 may be set by any authorized system operator indicating whether the in-vehicle network gateway 136 will execute hardware acceleration for protocol translation, including message filtering, pre-defined header lookup, and message aggregation, these three acceleration aspects is enabled or disabled separately within the configuration settings 606).

As per claim 22

Rajan teaches the system of claim 21, wherein the configuration circuit is further structured to selectively configure which of one or more portions of the first network interface circuit, the translation circuit, and/or the second network interface circuit are defined by the Ethernet switch and/or the CEG (Rajan see para 0036, 0039, in-vehicle network gateway 136 will execute hardware acceleration for protocol translation, including message filtering, pre-defined header lookup, and message aggregation, these three acceleration aspects is enabled or disabled separately within the configuration settings 606, message filter 210 determines which of the messages to pass for hardware accelerated protocol translation and which to discard or ignore, the translation table 212 defined for the Ethernet interface has an index input that receives

a table index value and for each value of the index input, stores a pointer to a pre-defined Ethernet header).

As per claim 23

Rajan teaches the system of claim 20, wherein the first network interface circuit is defined by the Ethernet switch and the second network interface circuit is defined by the CEG (Rajan see para 0041, the gateway circuitry 802 includes an Ethernet switch 804 with ports 806 and 808. The Ethernet switch 804 connects through Media Access Control circuitry MAC 810 to a system bus 812, bus transceiver circuitry 820 includes CAN bus transceivers connected to the CAN interface 138).

As per claim 24

Rajan teaches the system of claim 23, wherein the translation circuit is defined by the CEG (Rajan see para 0042, the CPUs 814 implement all or part of the control circuitry 214 , while the RAM 816 implement the memory system and store the routing table 211, translation table 212, pre-defined headers 218-1-218-*n*, the software protocol stack 604, and provide general purpose memory storage for the in-vehicle network gateway 136, the hardware acceleration circuitry 818 include, buffers 822 for storing messages, message filters 824, the hash circuitry 826, and the routing tables 828 for determining the destination interface as an alternative implementation to routing tables stored in the RAM 816).

As per claim 25

Rajan teaches the system of claim 23, wherein the translation circuit is defined by the Ethernet switch (Rajan see para 0044, 0051 the in-vehicle network gateway 136 use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to the CAN or other interface, the gateway circuitry 802 shapes the switch queues to match the rate of the target network or bus interface, in step 3 the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame).

***Claim Rejections - 35 USC § 103***

9. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

10. Claims 15 and 16 are rejected under AIA 35 U.S.C. 103 as being unpatentable over Rajan et al. (U.S. PGPub 2019/0141133) in view of Kim et al. (U.S. PGPub 2018/0072250).

As per claims 15

Rajan teaches the method of claim 13 yet fails to teach further comprising: generating the configuration command value external to the vehicle; and transmitting the configuration command value to the CND.

In a similar field of endeavor Kim teaches

further comprising: generating the configuration command value external to the vehicle, and transmitting the configuration command value to the CND (Kim, see para 0084, 0098, 0099, an external diagnostic apparatus 640 may be connected to the CAN-based vehicle network by a connector with the gateway 600 as the CND, and perform diagnostic procedures for the plurality of end nodes 511 to 516 belonging to the CAN-based vehicle network, external diagnostic apparatus is connected to the gateway the external diagnostic apparatus generate a diagnostic initiation message indicating that a diagnostic procedure by the external diagnostic apparatus is to be performed at S702, diagnostic initiation message as configuration command may include at least one information element as value an an identifier of the target end node to be diagnosed by the external diagnostic apparatus and Diagnostic type value indicating the type of the diagnostic procedure such as software update procedure to be performed by the external diagnostic apparatus, diagnostic time indicating performance time of the diagnostic procedure to be performed by the external diagnostic apparatus and sent the diagnostic message to the gateway 600).

It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to combine teaching of Rajan with the teaching of Kim as doing so would provide an efficient method for performing diagnostic procedures for vehicular network containing nodes using multiple different kind of network protocols communication using a common gateway from an internal and external source (Kim see para 0030).

As per claim 16

Rajan teaches the method of claim 12 yet fails to teach further comprising: generating a test command value external to the vehicle; transmitting the test command value to the CND; and responsive to the test command value, executing a test procedure involving a device on an Ethernet based network and/or on a CAN based network .

In a similar field of endeavor Kim teaches further comprising: generating a test command value external to the vehicle; transmitting the test command value to the CND (Kim, see para 0097, 0098, as shown in fig.6 and 7, the external diagnostic apparatus connected to the gateway 600 as CND generate a diagnostic initiation message indicating that a diagnostic procedure by the external diagnostic apparatus is to be performed with an identifier of the target end node to be diagnosed by the external diagnostic apparatus, an identifier included in the diagnostic initiation message may indicate all of end nodes belonging to the vehicle network, diagnostic type indicates the type of the diagnostic procedure with diagnostic time indicates performance time);

and responsive to the test command value, executing a test procedure involving a device on an Ethernet based network and/or on a CAN based network (Kim see para 0099, 0103, gateway 600 transmit the diagnostic initiation message in a broadcast manner, internal diagnostic apparatus may receive the diagnostic initiation message and confirm that the diagnostic procedure of the external diagnostic apparatus is to be performed based on the received diagnostic initiation message, the internal diagnostic apparatus may stop its diagnostic procedure for the target end node when it is possible to stop its diagnostic procedure for the target end node, if the diagnostic procedure for the target end node is stopped or the diagnostic procedure for the target end node is

completed, the internal diagnostic apparatus may generate a diagnostic response message with an identifier of the target end node diagnosed by the internal diagnostic apparatus, diagnostic type , diagnostic result indicates a result of the diagnostic procedure performed by the internal diagnostic apparatus).

It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to combine teaching of Rajan with the teaching of Kim as doing so would provide an efficient method for performing diagnostic procedures for vehicular network containing nodes using multiple different kind of network protocols communication using a common gateway from an internal and external source (Kim see para 0030).

## ***Conclusion***

11. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. This includes:

U.S. PGPub 2018/0286152 Apparatus and method for converting protocol by type of data and vehicle system.

U.S. PGPub 2019/0020985 System and method for wireless interface selection and for communication and access control of subsystems, devices, and data in a vehicular environment

U.S. PGPub 2017/0251339 System and method for routing, mobility, application services, discovery, and sensing in a vehicular network environment.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to examiner Sanjoy Roy, whose telephone number is 571-270-0675. The examiner can normally be reached on Mon-Fri, 8am.-5pm. (EST).

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Nicholas Taylor can be reached on 571-272-3889. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/SANJOY ROY/
Examiner, Art Unit 2443

/NICHOLAS R TAYLOR/
Supervisory Patent Examiner, Art Unit 2443